

Instruction:

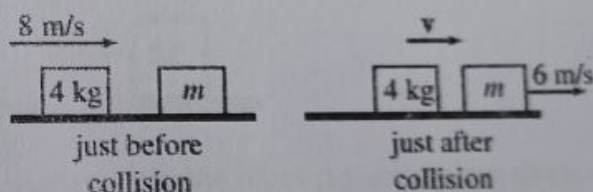
From the four alternative answers given, circle the one which you think is the best answer.

1. A spring of force constant k is stretched a certain distance. It takes twice as much work to stretch a second spring by half this distance. The force constant of the second spring is
- A. k
B. $2k$
C. $4k$
D. $8k$
E. $16k$

2. On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of the same mass that is initially at rest. After the collision, the first block moves at an angle θ to its initial direction with a speed $v/2$. The second block's speed after the collision is
- A. $\frac{\sqrt{3}}{4}v$
B. $\frac{1}{2}v$
C. $\frac{\sqrt{3}}{2}v$
D. $\frac{\sqrt{5}}{4}v$
E. $v + \frac{1}{2}v \cos \theta$

3. A system consists of two blocks which collide on a frictionless table as shown. Before collision, the block of mass m is at rest. What is the momentum and kinetic energy of the system after collision if the collision is elastic?

- A. $12 \text{ kg.m/s}, 16 \text{ J}$
B. $16 \text{ kg.m/s}, 32 \text{ J}$
C. $18 \text{ kg.m/s}, 64 \text{ J}$
D. $24 \text{ kg.m/s}, 128 \text{ J}$
E. $32 \text{ kg.m/s}, 128 \text{ J}$

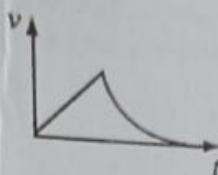


4. An object of mass m is traveling at constant speed v in a circular path of radius r . How much work is done by the centripetal force during one half of a revolution?
- A. Zero
B. $2\pi mv^2$
C. πmv^2
D. $\pi mv^2 r$
E. $2\pi mv^2 r$

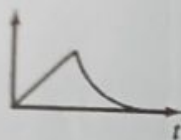
5. A ball of mass m traveling horizontally with velocity v strikes a massive vertical wall and rebounds back along its original direction with no change in speed. What is the magnitude of the impulse delivered by the wall to the ball?

- A. 0
B. mv
C. mv
D. $2mv$
E. $4mv$

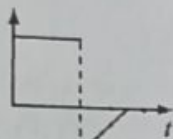
For an object traveling in a straight line, its velocity (v , in m/s) as a function of time (t , in s) is given by the following graph.



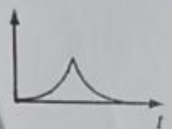
Which graphs best depict the object's momentum and kinetic energy?



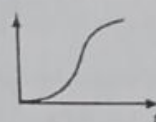
(I)



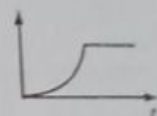
(II)



(III)



(IV)

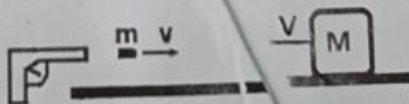


(V)

- A. I and II
- B. I and III
- C. II and III

- D. IV and V
- E. II and IV

7. A wooden block of mass M is moving at speed V in a straight line as shown.



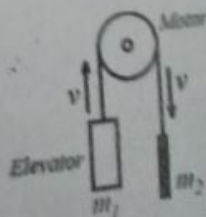
How fast would the bullet of mass m need to travel to stop the block (assuming that the bullet became embedded inside)?

- A. $mV/(m + M)$
- B. MV/m
- C. mV/M

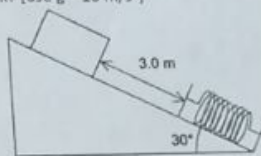
- D. $MV/(m + M)$
- E. $(m + M)V/m$

8. The diagram shows a lift system in which the elevator (m_1) is counterbalanced by a heavy weight (m_2). What power is provided by the motor to the system while the elevator is rising at a steady speed v ?

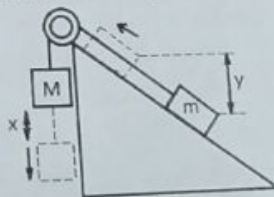
- A. $\frac{1}{2}mv^2$
- B. $\frac{1}{2}(m_1 - m_2)v^2$
- C. M_1gv
- D. $(m_1 - m_2)gv$
- E. $\frac{1}{2}(m_1 + m_2)v^2$



9. A block of mass 12 kg is released from rest and it slides down a 30° frictionless incline plane. It is stopped by a spring with a spring constant k as shown below. The initial distance between the point of release and the uncompressed spring is 3.0 m. After the block comes to rest momentarily, the spring is compressed by 1.3 m. What is the spring constant k ? [use $g = 10 \text{ m/s}^2$]
- 213 N/m
 - 250 N/m
 - 305 N/m
 - 350 N/m
 - 500 N/m



10. The figure shows mass m on an inclined plane connected by an inextensible string over a smooth pulley to a larger mass M . When the mass M falls through a distance x , the mass m moves up a vertical distance y at a constant speed. The friction between m and the inclined plane is F . What is the change in the mechanical energy?
- $F(x - y)$
 - Fy
 - $Mgx - mgy$
 - $(M - m)gx$
 - Fx

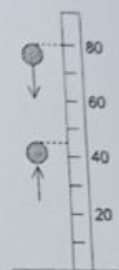


11. A car of mass 1000 kg has kinetic energy $4.5 \times 10^5 \text{ J}$. It comes to rest by a total constant braking force of 6000 N. What is its stopping distance?
- 37 m
 - 75 m
 - 150 m
 - 300 m
 - 100 m

12. Car X is twice as heavy as car Y and travels at half the speed of car Y. Compare the kinetic energies of the cars.
- Car X has half the kinetic energy of car Y
 - Car X has a quarter of the kinetic energy of car Y
 - Car X has twice the kinetic energy of car Y
 - Car X has four times the kinetic energy of car Y
 - Car X has the same kinetic energy as car Y

13. Three particles of mass 40 g each are at the points (7.07 cm, 0 cm), (-7.07 cm, 0 cm) and (0 cm, -7.07 cm) and a particle of mass 90 g is at the point (0 cm, 7.07 cm). What are coordinates of the center of mass of the system?
- (1.41 cm, 0 cm)
 - (0 cm, 0.909 cm)
 - (0 cm, 0 cm)
 - (0.786 cm, 0.72 cm)
 - (0 cm, 2.72 cm)

14. A solid rubber ball has a diameter of 8.0 cm. It is released from rest with the top of the ball at 80 cm above a horizontal surface. It falls vertically and bounces up to a maximum height of 45 cm as shown. If the kinetic energy of the ball when it strikes the surface is 0.75 J what is its kinetic energy just after it leaves the surface?
- A. 0.36 J
 B. 0.39 J
 C. 0.40 J
 D. 0.42 J
 E. 0.75 J



15. A small mass is released from rest at point P on the inner surface of a smooth hemisphere. Its speed at different points down the hemisphere are as indicated on the diagram below. At what point has the mass lost one fourth of its potential energy?
- A. Q
 B. R
 C. S
 D. Between R and S
 E. Between S and T

